

Latent Potential Estimation Framework for Traditional Trade FMCG Outlets Using Geospatial Intelligence and Censored Demand Reconstruction

Team: InferaX

1) Executive Summary

Traditional FMCG trade allocation strategies rely heavily on historical sales averages to distribute trade marketing budgets, coolers, and promotional investments. However, historical sales alone fail to represent true market opportunity because observed purchases are frequently constrained by operational limitations such as stockouts, cooler limitations, delivery bottlenecks, and retailer credit ceilings.

This project introduces a Latent Potential Estimation Framework designed to reconstruct uncapped outlet demand by combining data engineering, geospatial intelligence, environmental clustering, and conservative uncensoring methodologies. Rather than forecasting future sales directly, the framework estimates the theoretical maximum monthly purchase potential of each retail outlet under ideal operational conditions.

A Bronze → Silver → Gold Lakehouse architecture was implemented to ensure reproducibility and structured data processing. Legacy system artifacts including duplicate transaction retries, zero-volume ghost rows, pricing anomalies, and stale master-data labels were identified through reusable forensic validation checks and quarantined into a rejected-records store.

To enrich transactional data with external market context, geospatial POI signals were acquired using OpenStreetMap (OSM), OSMnx, and the Overpass API. Nearby schools, bus stops, hospitals, and similar catchment drivers were mapped to outlet-level buffers and converted into proxy demand features representing latent commercial intensity.

The final methodology leveraged environmental clustering and peer benchmarking to estimate uncapped demand ceilings. Top-performing outlets within comparable environmental clusters were treated as proxy indicators of achievable latent capacity, while conservative uncensoring logic was used to avoid unrealistic inflation and overestimation bias.

The resulting framework enables a transition from Historical-Based Allocation toward Potential-Based Allocation, allowing strategic prioritization of outlets with the largest latent growth opportunity rather than simply reinforcing historically high-performing outlets.

2) DATA FORENSICS & HYGIENE

Bronze → Silver → Gold Architecture

A structured Lakehouse-style analytical pipeline was implemented to ensure data reproducibility and controlled transformation across all processing stages.

Layer	Purpose
Bronze	Raw immutable ingestion of source files
Silver	Data quality validation, forensic cleaning, quarantining
Gold	Feature engineering and latent-demand enrichment

The Bronze layer preserved all source datasets exactly as received. The Silver layer applied reusable validation checks and forensic-cleaning logic, while the Gold layer generated model-ready outlet-level analytical features enriched with behavioral, operational and geospatial intelligence.

Data Forensics and Legacy System Artifacts

The transactional dataset contained 2.376 million outlet-level records across 36 months of FMCG operations. Initial forensic analysis revealed that while overall data integrity was strong (99.8% valid records), approximately 10.2% of transactions exhibited patterns inconsistent with unconstrained retail demand.

Several categories of operational artifacts were identified:

Artifact Type	Count	Interpretation
Negative Volumes	4,753	ERP reversals / returns
Zero-Volume Rows	100	Ghost system entries
Pricing Anomalies	236,818	Premium SKU_09 category
Volume Ceiling Patterns	1.2M+ outlet-months	Demand censoring evidence

Negative Volume Transactions

Negative-volume transactions represented operational reversals rather than negative demand. These records were geographically concentrated in the Western region and appeared consistently throughout the 36-month timeline, strongly suggesting ERP reverse-posting behavior rather than true commercial decline.

These records were removed from downstream demand estimation because including them would artificially suppress latent-demand signals. However, all removed records were quarantined into a rejected-records audit layer with explicit rejection codes for forensic traceability.

Zero-Volume Ghost Records

Zero-volume records represented incomplete transaction posts, failed mobile sync operations, or reconciliation placeholders. These records generated undefined revenue-per-liter calculations and introduced invalid downstream feature-engineering behavior.

Such records were excluded from analytical pipelines while preserving audit visibility through the rejected-records store.

The SKU_09 Discovery

One of the most important forensic findings involved SKU_09, which initially appeared as a statistical pricing anomaly due to extremely high revenue-per-liter values. However, deeper investigation revealed that SKU_09 represented a structurally different premium/concentrate product category rather than corrupted data.

Instead of removing these records as outliers, they were retained and treated as a separate commercial product segment. This decision preserved critical high-margin demand signals that would otherwise have been lost through naive statistical cleaning.

This demonstrated a key forensic principle applied throughout the project:

statistical anomaly does not necessarily imply data error.

3) SPATIAL INTELLIGENCE & POI ENRICHMENT

Why External Spatial Intelligence Was Required

Transactional history alone cannot fully explain latent demand because observed sales are operationally constrained. Two outlets with identical historical volumes may possess fundamentally different commercial opportunity depending on their surrounding environmental context.

To reconstruct uncaptured demand landscapes, external geospatial intelligence was integrated into the analytical pipeline using OpenStreetMap (OSM) data.

POI Data Acquisition Strategy

External Point-of-Interest (POI) data was acquired using:

- OpenStreetMap (OSM),

- OSMnx,
- and the Overpass API.

District-level spatial queries were used to retrieve:

- schools
- bus stops
- hospitals

The acquired POI datasets were cached locally as GeoJSON files to:

- avoid repeated API calls
- improve reproducibility
- reduce Overpass API load
- and accelerate iterative analysis workflows

POIs as Proxy Demand Signals

POIs were treated as proxy indicators of human movement and commercial footfall rather than direct demand measurements.

POI Type	Commercial Interpretation
Schools	Youth-driven consumption demand
Bus Stops	High transient commuter footfall
Hospitals	Stable continuous daily traffic

These environmental signals reveal latent commercial opportunity unavailable in transactional data alone. An outlet near a major school or bus stand experiences structurally higher demand exposure than an isolated rural outlet, even if current sales volumes appear similar.

Spatial Mapping Methodology

Outlet coordinates were converted into GeoPandas geospatial point layers using EPSG:4326 coordinates and projected into meter-based coordinate systems for accurate spatial operations.

A 500-meter catchment buffer was created around each outlet to approximate the local pedestrian-accessible commercial zone. Spatial joins were then performed between outlet buffers and POI datasets to calculate localized environmental-demand density.

The resulting spatial features included:

- Nearby_Schools,
- Nearby_Bus_Stops,
- Nearby_Hospitals,
- and aggregate spatial demand indicators.

These geospatial features enabled the analytical pipeline to quantify environmental commercial intensity independent of current sales behavior.

4) LATENT DEMAND ESTIMATION & CAUSAL BASE LOGIC

The Censoring Problem

The central analytical challenge was that true outlet demand was not directly observable. Instead, observed transactional sales represented censored behavior constrained by operational ceilings.

The relationship was modeled as:

$$Y = \min(Y^*, C)$$

Where:

- Y = observed sales,
- Y^* = latent true demand,
- C = operational ceiling.

Operational ceilings included:

- cooler capacity,
- distributor allocation policies,
- sales-representative visit frequency,
- and retailer credit limits.

This meant that historical averages alone systematically underestimated true outlet capacity.

Environmental Clustering Methodology

Traditional sales-based clustering was intentionally avoided because constrained outlets cluster artificially with low-demand outlets, masking latent opportunity.

Instead, environmental clustering grouped outlets based on:

- POI density,
- cooler infrastructure,
- SKU diversity,
- and operational context.

K-Means clustering was applied to standardized environmental features, creating peer groups representing similar commercial ecosystems rather than similar historical sales histories.

This allowed outlets operating under similar demand landscapes to be benchmarked against one another independent of historical operational constraints.

Peer Benchmarking and Cluster Ceilings

Within each environmental cluster, top-performing outlets were treated as indicators of achievable uncapped commercial behavior.

Rather than using maximum values directly, cluster ceilings were estimated using upper-percentile peer performance to reduce sensitivity to outlier inflation.

This peer benchmarking logic assumed:

if one outlet can achieve high volumes under a given environmental context, comparable outlets operating below that ceiling likely possess uncaptured latent demand.

Conservative Uncensoring Strategy

A conservative uncensoring strategy was adopted to avoid unrealistic inflation of estimated demand.

Instead of aggressively extrapolating latent capacity, only controlled peer-gap uplift was transferred into final potential estimates. This balanced:

- opportunity identification
- operational realism
- and overestimation risk mitigation.

This approach prioritized business defensibility over speculative growth inflation.

Strategic Business Impact

The resulting framework supports:

- Potential-Based Allocation,
- targeted cooler deployment,
- optimized distributor credit allocation,
- and precision trade-marketing investment.

Rather than rewarding historically unconstrained outlets exclusively, the framework identifies constrained outlets with structurally high latent commercial opportunity.

5) GENAI TRANSPARENCY LOG & CONCLUSION

GenAI Transparency Log

Generative AI tools including:

- ChatGPT,
- GitHub Copilot,
- Gemini

were used throughout the 36-hour competition workflow as engineering accelerators and analytical reasoning assistants.

LLMs assisted with:

- brainstorming censored-demand methodologies,
- refining spatial intelligence workflows,
- generating reusable data-engineering utilities,
- debugging geospatial processing pipelines,
- validating clustering strategies,

However, all AI generated outputs were critically reviewed, validated, and iteratively refined against actual dataset behavior and business logic before integration into the final analytical framework.

The team maintained full human oversight across:

- methodology design
- feature engineering
- data validation
- and business interpretation stages

This ensured that Generative AI functioned as an engineering productivity enhancer rather than a decision making substitute.

